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A																							
B																							
C																							
D																							
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F																							
UniSec Medium Voltage Switchgear PT MULTIMAS NABATI ASAHAN MVD-27 20kV Tank Farm SCHEMATIC DIAGRAM SBC-W OUTGOING H02																							
				Date Prep.		15.11.2018		PROJECT TITLE				ABB		EPMV		COVER SHEET		Scale		NTS		= H02	
				Prepared		YR		MV Switchgear for MNA Serang Plant				PT. ABB Sakti Industri		MVD-27 20kV Tank Farm SCHEMATIC DIAGRAM SBC-W OUTGOING		Customer		PT Multimas Nabati Asahan		& 306			
00		Issued for Approval		15.01.19		YR										Checked		AW		Drawing Nbr.		18.042-MNA-306.08	
Ind.		Revision		Date		Name		Approved		AI		Based on: BE318042				Customer Doc. Nbr.				Cont.		27	
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SHEET INDEX

PAGE	PAGE NAME	REVISION		
==MVD-27=H02&306/1	COVER SHEET	00		
==MVD-27=H02&306/2	Index of Sheet	00		
==MVD-27=H02&306/3	REFERENCE DESIGNATIONS	00		
==MVD-27=H02&306/4	REFERENCE DESIGNATIONS	00		
==MVD-27=H02&306/5	REFERENCE DESIGNATIONS	00		
==MVD-27=H02&306/6	REFERENCE DESIGNATIONS	00		
==MVD-27=H02&306/7	DISTRIBUTION OF AUXILIARY CIRCUITS	00		
==MVD-27=H02&306/8	AUXILIARY SUPPLY CIRCUIT	00		
==MVD-27=H02&306/9	CIRCUIT BREAKER CONTROL	00		
==MVD-27=H02&306/10	AUXILIARY MOTOR & RELAY	00		
==MVD-27=H02&306/11	DIGITAL INPUT/OUPUT RELAY	00		
==MVD-27=H02&306/12	SPARE CONTACT	00		
==MVD-27=H02&306/13	CURRENT TRANSFOMER	00		
==MVD-27=H02&306/14	VOLTAGE TRANSFORMER	00		
==MVD-27=H02&306/15	LV DOOR LAYOUT	00		
==MVD-27=H02&306/16	LV BOX LAYOUT	00		
==MVD-27=H02&306/17	PART LIST	00		
==MVD-27=H02&306/18	Terminal diagram ==MVD-27=H02+-XDA			
==MVD-27=H02&306/19	Terminal diagram ==MVD-27=H02+-XDB1			
==MVD-27=H02&306/20	Terminal diagram ==MVD-27=H02+-XDC1			
==MVD-27=H02&306/21	Terminal diagram ==MVD-27=H02+-XDC2			
==MVD-27=H02&306/22	Terminal diagram ==MVD-27=H02+-XDI			
==MVD-27=H02&306/23	Terminal diagram ==MVD-27=H02+-XDI2			
==MVD-27=H02&306/24	Terminal diagram ==MVD-27=H02+-XDI3			
==MVD-27=H02&306/25	Terminal diagram ==MVD-27=H02+-XDI6			
==MVD-27=H02&306/26	Terminal diagram ==MVD-27=H02+-XDV			
==MVD-27=H02&306/27	Terminal diagram ==MVD-27=H02+-XDX			

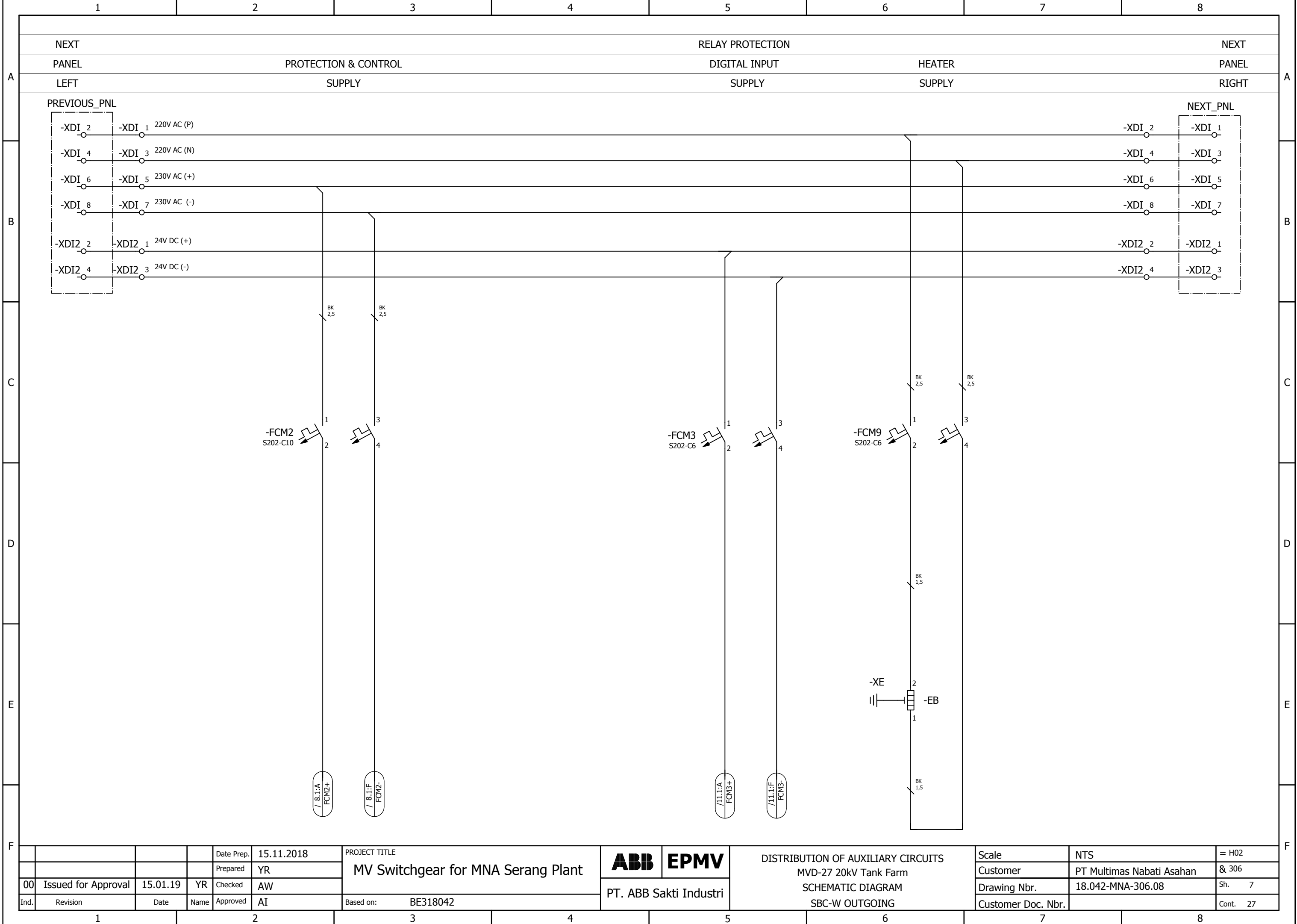
			Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	Index of Sheet MVD-27 20kV Tank Farm SCHEMATIC DIAGRAM SBC-W OUTGOING	Scale	NTS	= H02	
			Prepared	YR							Customer	PT Multimas Nabati Asahan
00	Issued for Approval	15.01.19	YR	Checked		AW	PT. ABB Sakti Industri			Drawing Nbr.	18.042-MNA-306.08	Sh. 2
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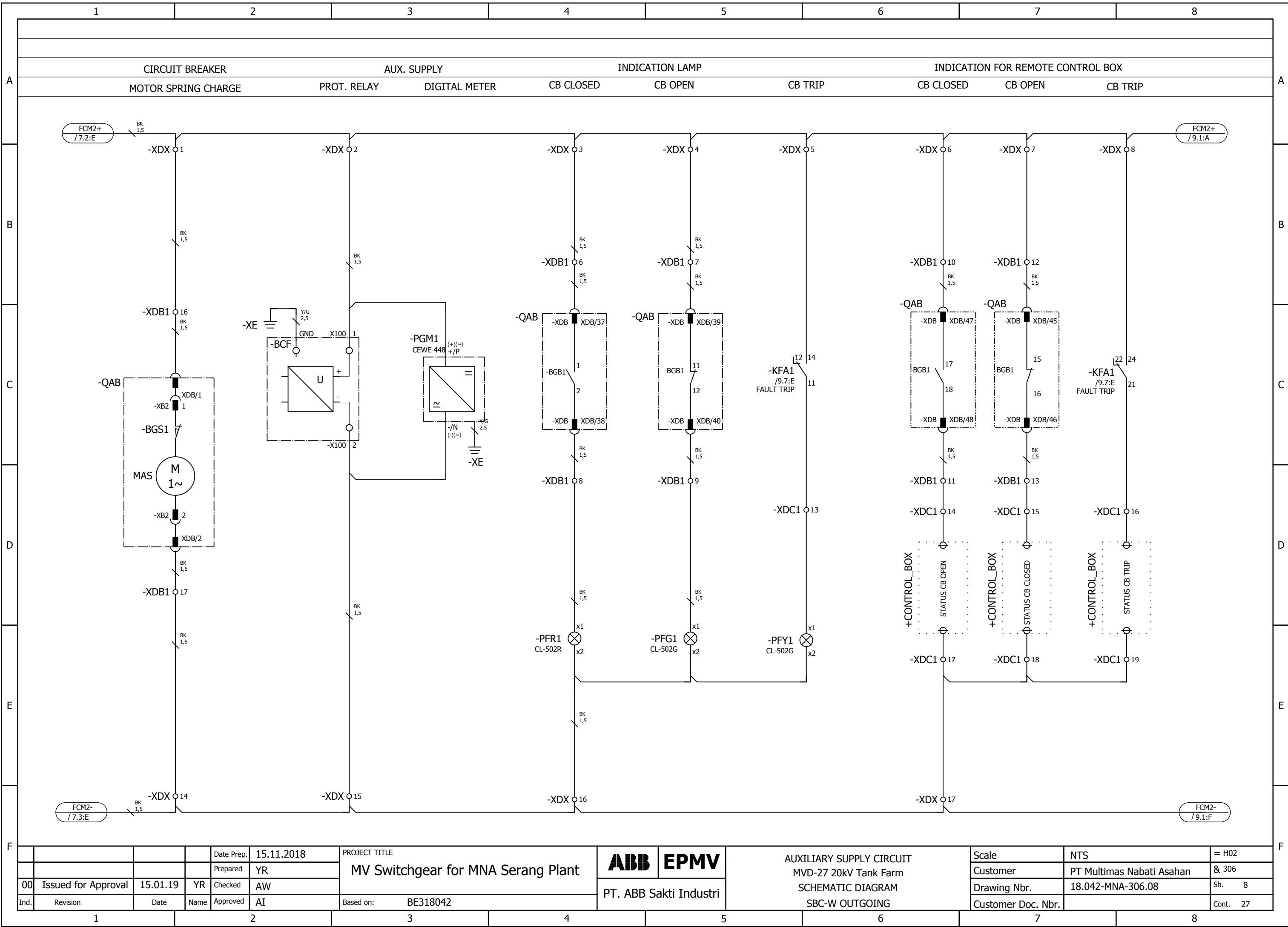
A	-TFA	ACTIVE POWER TRANSDUCERS	-XDI1	TERMINAL BLOCK FOR INTERCONNECTIONS OF CONTROL CIRCUITS	A
	-TFC	CURRENT TRANSDUCERS		-XDI2	
B	-TFF	FREQUENCY TRANSDUCERS	-XDI3	TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR SWITCH-DISCONNECTOR OPERATION	B
	-TFJ	ACTIVE ENERGY TRANSDUCERS	-XDI4	TERMINAL BLOCK FOR INTERCONNECTIONS OF A.C. AUXILIARY CIRCUITS	
	-TFK	REACTIVE ENERGY TRANSDUCERS	-XDI6	TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS) OF VOLTAGE CIRCUITS	
	-TFM	MULTIFUNCTION TRANSDUCERS			
	-TFP	POWER-FACTOR TRANSDUCERS			
	-TFQ	REACTIVE POWER TRANSDUCERS	-XDI7	TERMINAL BLOCK FOR INTERCONNECTION OF MOD-BUS	
	-TFS	SIGNAL CONVERTERS	-XDI8	TERMINAL BLOCK FOR INTERCONNECTIONS OF CURRENT CIRCUITS	
	-TFV	VOLTAGE TRANSDUCERS	-XDI9	TERMINAL BLOCK FOR SPECIAL USE	
	-WA	BUSBARS	-XDJ	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF DISCONNECTORS -QBD	
	-WBC	POWER CABLES	-XDJ1	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR -QBS	
C	-WE	EARTHING CABLES FOR L.V. INSTRUMENT COMPARTMENT.	-XDM	SEALABLE TERMINAL BLOCK FOR MEASUREMENT	C
	-WGC	CONTROL CABLES	-XDS	CURRENT SOCKETS	
	-WGS	COMMUNICATION CABLES (SHIELDED TWISTED PAIRS)	-XDT	TERMINAL BLOCK FOR POSITION CONTACTS OF TRUCK AND OF CONNECTOR -XDB	
	-WH	OPTICAL FIBRES	-XDU	CONNECTOR FOR ISOLATION OF CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK	
	-XDA	TERMINAL BLOCK FOR CIRCUITS OF CURRENT TRANSFORMERS	-XDU1	TERMINAL BLOCK FOR CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK	
	-XDA1	CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L1	-XDV	TERMINAL BLOCK FOR CIRCUITS OF VOLTAGE TRANSFORMERS	
	-XDA2	CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L2	-XDV1	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L1	
	-XDA3	CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L3	-XDV2	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L2	
	-XDA5	CONNECTOR FOR CIRCUITS OF NEUTRAL (RESIDUAL) CURRENT TRANSFORMER	-XDV3	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L3	
	-XDB	CONNECTOR FOR ISOLATION OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK) AUXILIARY CIRCUITS	-XDV4	CONNECTOR FOR CIRCUITS OF ANTIFERRORESONANCE RESISTOR	
D	-XDB1	CONNECTOR FOR AUXILIARY CIRCUITS OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK)	-XDX	SUPPORT TERMINAL BLOCK	D
	-XDB2	CONNECTOR FOR ADDITIONAL AUXILIARY CIRCUITS OF CIRCUIT-BREAKER	-XE1	EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, LEFT SIDE	
	-XDB10...-XDB89	CONNECTOR FOR CIRCUIT-BREAKER INTERNAL CIRCUITS	-XE2	EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, RIGHT SIDE	
	-XDB7	CONNECTOR FOR TRUCK POSITION CONTACTS	-XE5	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, VOLTAGE INDICATOR SIDE	
	-XDB91	CONNECTOR FOR OPENING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR	-XE6	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, REAR BOTTOM SIDE	
	-XDB92	CONNECTOR FOR CLOSING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR	-XE7	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, LEFT SIDE	
	-XDB93	CONNECTOR FOR POSITION SWITCHES OF SWITCH-DISCONNECTOR	-XE8	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, FRONT BOTTOM SIDE	
	-XDB95	CONNECTOR FOR CIRCUITS OF EARTHING SWITCH LOCKING MAGNET	-XE9	EARTHING TERMINAL BLOCK FOR TRANSFORMERS IN BUSBAR COMPARTMENT	
	-XDC	CUSTOMER TERMINAL BLOCK			
	-XDC1	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF CIRCUIT BREAKER			
E	-XDC2	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR			E
	-XDC3	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF PROTECTION RELAY AND MULTIFUNCTION UNIT			
	-XDC4	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF VOLTAGE TRANSFORMERS			
	-XDC5	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF MINIATURE CIRCUIT BREAKER			
	-XDE	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE			
	-XDE1	TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE			
	-XDE2	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2			
	-XDE3	TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2			
	-XDF	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF FORCED COOLING VENTILATORS			
	-XDF1	CONNECTOR FOR ISOLATION OF CIRCUITS OF FRONT VENTILATOR			
F	-XDF2	CONNECTOR FOR REAR VENTILATOR CIRCUITS			F
	-XDF3	CONNECTOR FOR FRONT VENTILATOR CIRCUITS			
	-XDH	TERMINAL BLOCK FOR VARIOUS CIRCUITS (HEATERS, POSITION SWITCHES DETECTING INTERNAL ARCING, ETC.)			
	-XDI	TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS)			

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant		ABB	EPMV	REFERENCE DESIGNATIONS MVD-27 20kV Tank Farm SCHEMATIC DIAGRAM SBC-W OUTGOING	Scale	NTS	= H02
				Prepared	YR						Customer	PT Multimas Nabati Asahan	& 306
00	Issued for Approval	15.01.19	YR	Checked	AW						Drawing Nbr.	18.042-MNA-306.08	Sh. 6
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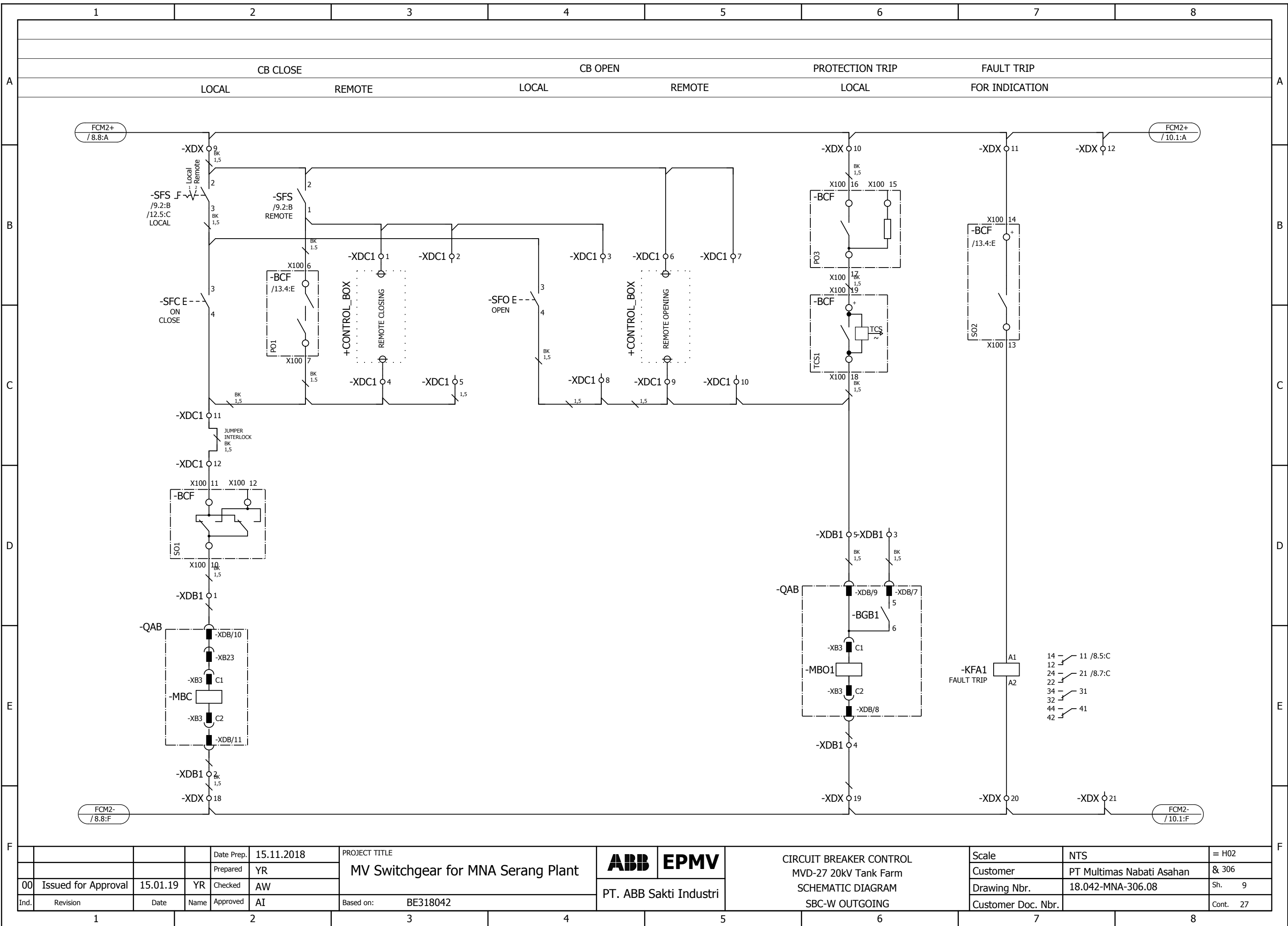
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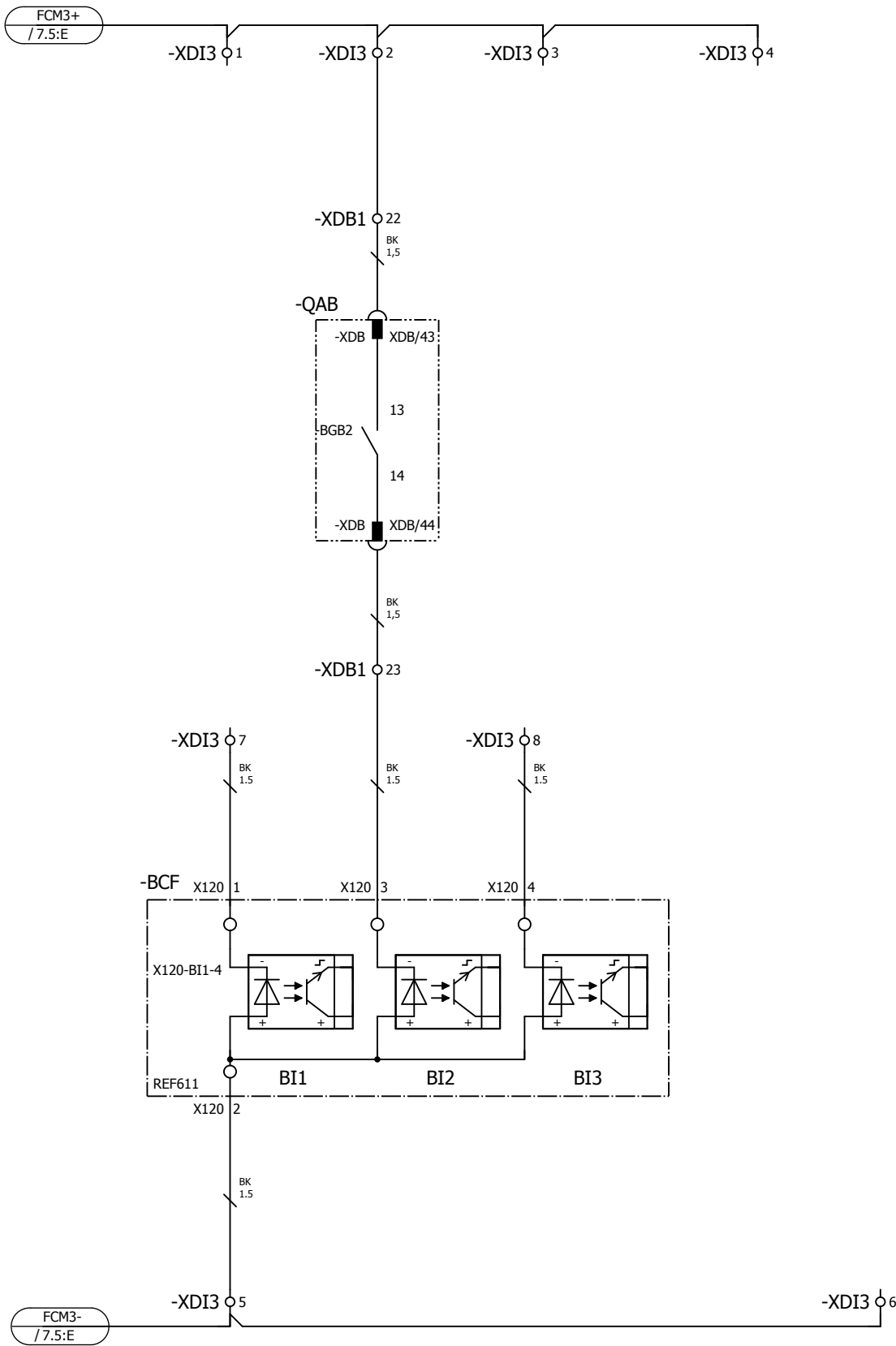


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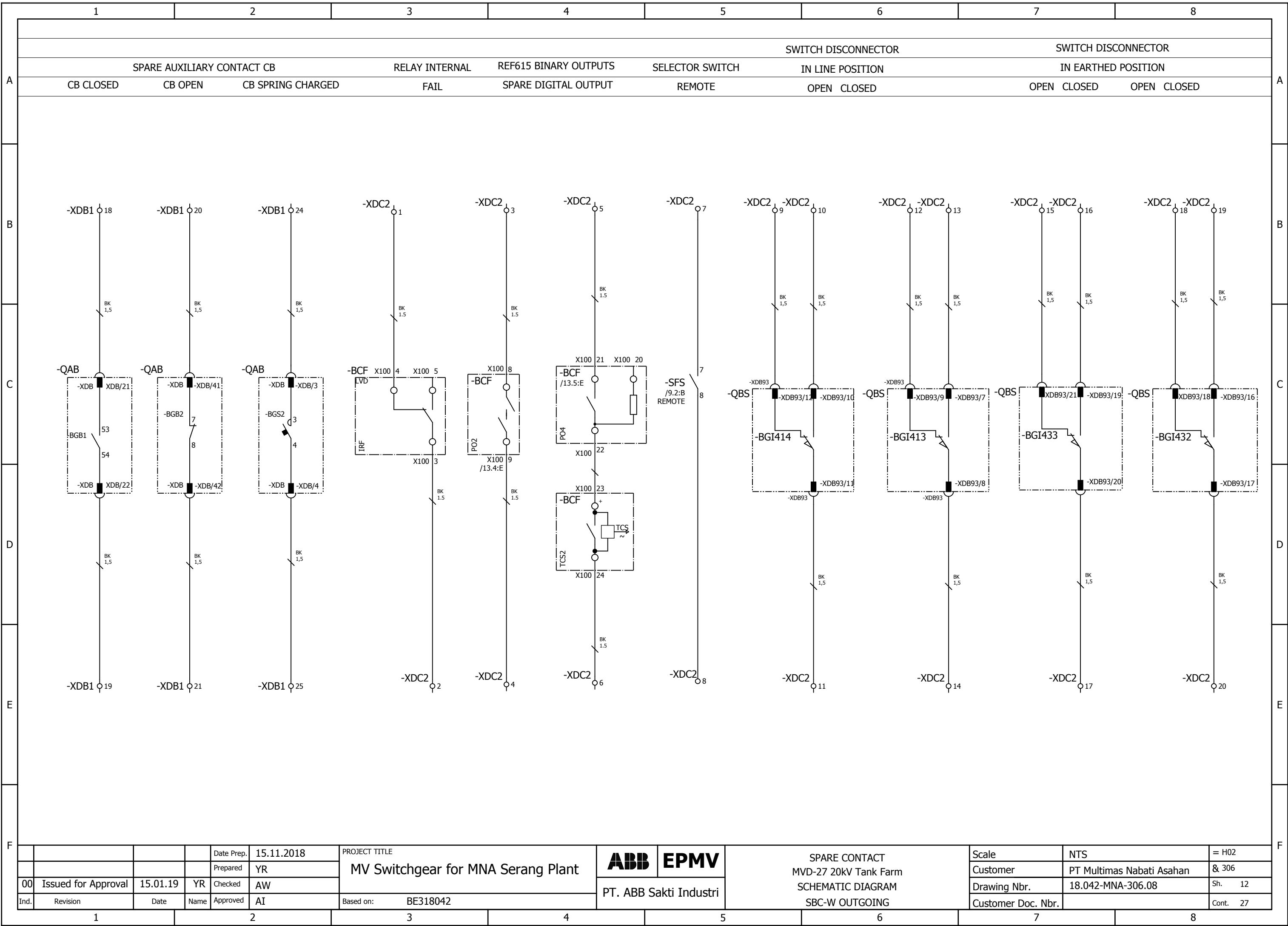
				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	AUXILIARY MOTOR & RELAY MVD-27 20kV Tank Farm SCHEMATIC DIAGRAM SBC-W OUTGOING	Scale	NTS	= H02
				Prepared	YR					Customer	PT Multimas Nabati Asahan	& 306
00	Issued for Approval	15.01.19	YR	Checked	AW		Drawing Nbr.	18.042-MNA-306.08		Sh. 10		
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						PT. ABB Sakti Industri						

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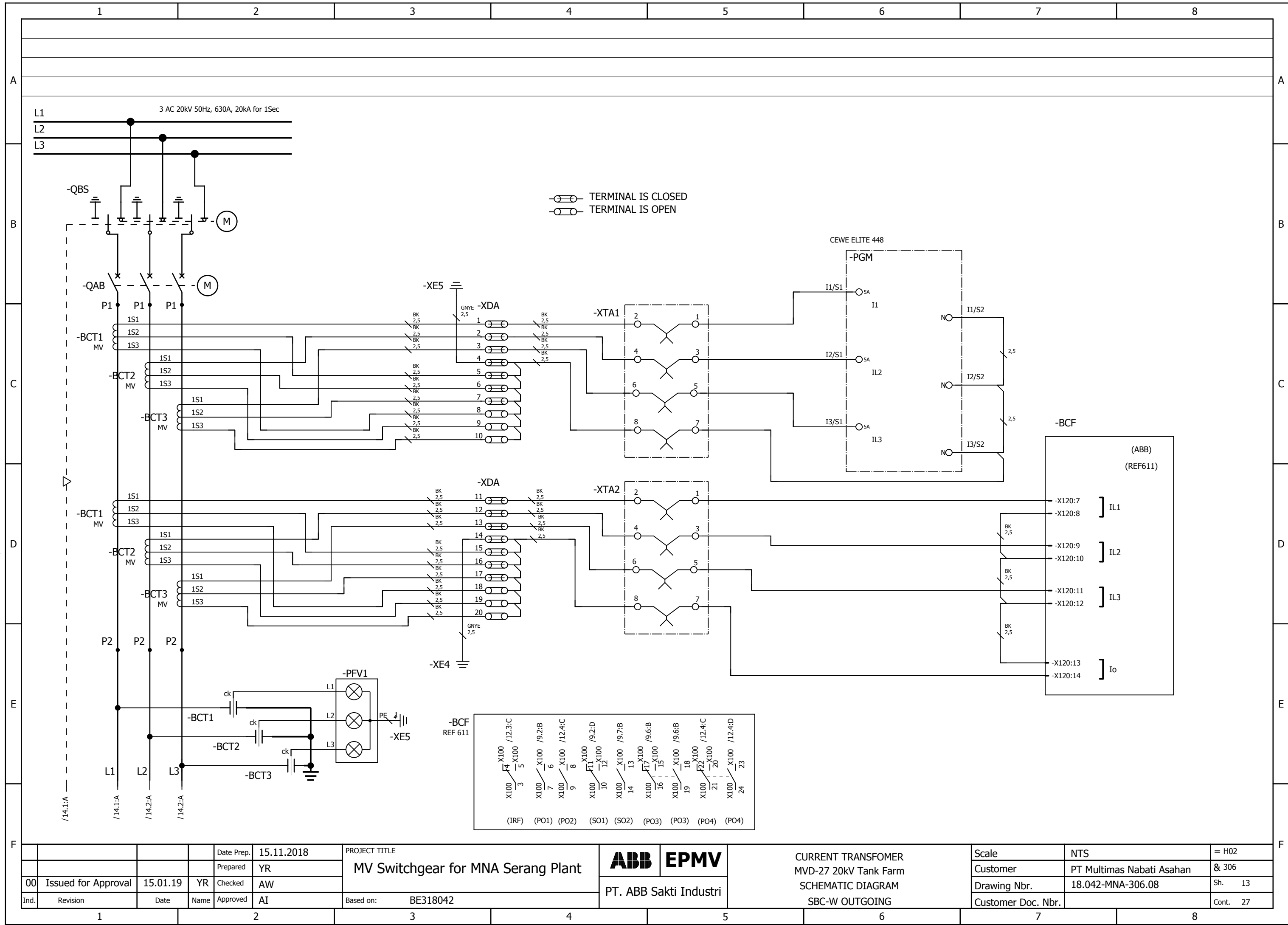


00	Issued for Approval	15.01.19	YR	Date Prep.	15.11.2018	PROJECT TITLE	MV Switchgear for MNA Serang Plant	ABB EPMV	DIGITAL INPUT/OUPUT RELAY MVD-27 20kV Tank Farm SCHEMATIC DIAGRAM SBC-W OUTGOING	PT. ABB Sakti Industri	Scale	NTS	= H02
				Prepared	YR						Customer	PT Multimas Nabati Asahan	& 306
				Checked	AW						Drawing Nbr.	18.042-MNA-306.08	Sh. 11
				Approved	AI						Customer Doc. Nbr.		Cont. 27
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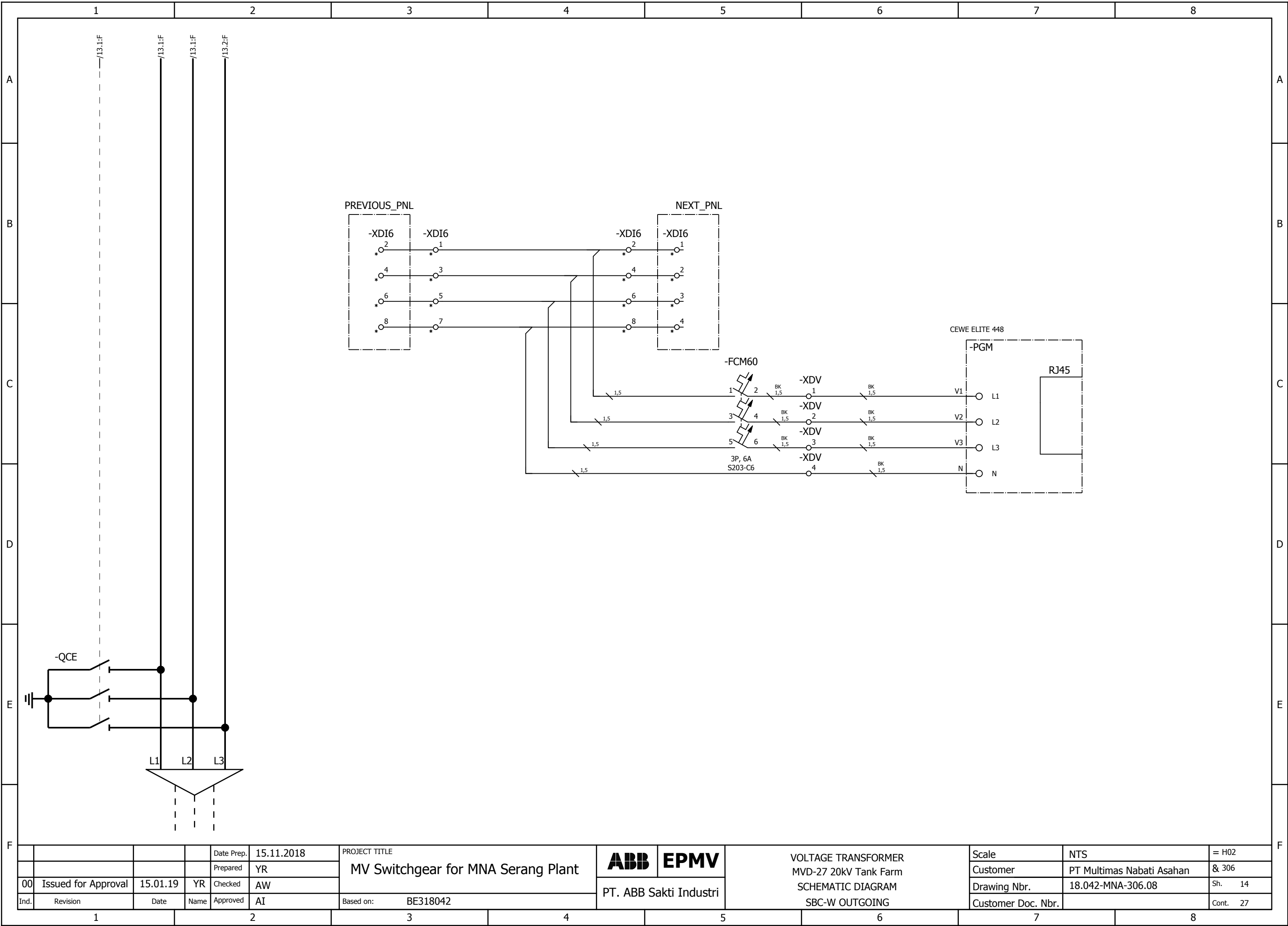
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				Prepared	YR									Customer	PT Multimas Nabati Asahan	& 306
				Checked	AW									Drawing Nbr.	18.042-MNA-306.08	Sh. 14
				Approved	AI									Customer Doc. Nbr.		Cont. 27

No.	EQUIPMENT DESCRIPTION	ENGRAVING
1	-SFS (LOCAL-OFF-REMOTE SELECTOR SWITCH, K&N)	-SFS, LOCAL-OFF-REMOTE
2	-SFC (PUSH BUTTON CB CLOSE, GREEN, ABB)	-SFC, CB CLOSE
3	-SFO (PUSH BUTTON CB OPEN, RED, ABB)	-SFO, CB OPEN
4	-PFY1 (LAMP INDICATION FOR CB TRIP, YELLOW, ABB)	-PFY1, CB TRIP
5	-PFR1 (LAMP INDICATION FOR CB CLOSE, RED, ABB)	-PFR1, CB CLOSE
6	-PFG1 (LAMP INDICATION FOR CB OPEN, GREEN, ABB)	-PFG1, CB OPEN
7	-PGJ (DIGITAL MEASUREMENT, CEWE 448)	-PGJ, DIGITAL MEASUREMENT
8	-BCF (FEEDER PROTECTION RELAY, REF615J, ABB)	-BCF, RELAY PROTECTION
9	-XTA1, XTA2 (TEST BLOCK , YONGSUNG)	-XTA, CURRENT TEST BLOCK

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			Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	LV BOX LAYOUT MVD-27 20kV Tank Farm SCHEMATIC DIAGRAM SBC-W OUTGOING	Scale	NTS	= H02
			Prepared	YR					Customer	PT Multimas Nabati Asahan	& 306
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Summarized parts list															
IDABB_F02_005-IDABB															
Carry over: 0.00															
TAG ID		DESCRIPTION				MANUFACTUR/BRAND		PART NUMBER		QUANTITY		REMARKS			
-BCF		Feeder Management Relay REF611HBAABC1NN11G				ABB		REF611HBAABC1NN11G		1		/8.2:C			
-EB		HEATER 230VAC, 50W				SANYANG		CDXN010033P0009		1		/7.6:E			
-FCM2		MCB S202-C10 MCB. 2P, 10A				ABB		2CDS252001R0104		1		/7.2:C			
-FCM3;-FCM9		MCB S202-C6 MCB. 2P, 6A				ABB		2CDS252001R0064		2		/7.5:C;/7.6:C			
-FCM60		MCB S203-C6 MCB. 3P, 6A				ABB		2CDS253001R0064		1		/14.5:C			
-KFA1		Auxiliary Relay 230AC 4COS				PHOENIX		1YRA3DL266230001		1		/9.7:E			
-PFG1		PILOT LIGHT GREEN #CL-523G#230VAC CL-523G				ABB		1SFA619402R5232		1		/8.5:E			
-PFR1		PILOT LIGHT RED #CL-523R#230VAC CL-523R				ABB		1SFA619402R5231		1		/8.4:E			
-PFY1		PILOT LIGHT YELLOW #CL-523Y#230VAC CL-523Y				ABB		1SFA619402R5233		1		/8.6:E			
-PGM1		POWER METER CEWE 448 Class : 0.5s; CT Input : 1A; VT Input : 57.7-240V (L-N); Aux. Voltage: 80-300VAC/DC				CEWE		1YRA4EA0350001		1		/8.3:C			
-SFC		PUSH BUTTON, YW1B-M1E10-G GREEN COLOUR				IDEC		1YRA3DA14700001		1		/9.2:B			
-SFO		PUSH BUTTON, YW1B-M1E10-R RED COLOUR				IDEC		1YRA3DA14700002		1		/9.4:B			
-SFS		LOCAL/REMOTE Selector Switch Type : S0-CA10-A723-600-E; 4NC (for LOCAL Position) & 4NO (for REMOTE Position)				K&N		1YRA3DA17940011		1		/9.2:B			
-XTA1;-XTA2		Current Test Block Rated Current : 10A, 250V; Mounting : Flush Mounting				YONGSUNG		YSCTT-04C		2		/13.4:C;/13.4:D			
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